Formalizing Fermat, Lecture 11

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EPSRC TCC, 14th May 2026

Recap of last time

Last time we saw several fairly deep results about cohomology of local Galois groups, and then several very deep (but very old) results about the cohomology of global Galois groups.

In particular, say k is a finite field of characteristic $\ell > 2$, if S is a finite set of finite places of the number field K containing all primes above ℓ, and say M is a finite k-module with an action of $G_S := \mathrm{Gal}(K^S/K)$.

Say $\mathcal{L}$ is a collection of subspaces $L_P \subseteq H^1(\mathrm{Gal}(\overline{K}_P/K_P), M)$ for $P \in S$, and $H^1_{\mathcal{L}}(G_S, M)$ is the elements in $H^1(G_S, M)$ which locally restrict to L_P for all $P \in S$.

Say $\mathcal{L}^\perp$ is the collection of the annihilators $L_P^\perp \subseteq H^1(\mathrm{Gal}(\overline{K}_P/K_P), M^*)$.

Then two deep theorems (Poitou–Tate LES, global Euler characteristic) gave us an easy-to-calculate formula for $\dim_k(H^1_{\mathcal{L}}(G_S, M)) - \dim_k(H^1_{\mathcal{L}^\perp}(G_S, M^*))$.

Recap of last time

We then saw that if G was any profinite group and $R^\square$ was a universal lifting ring for a mod ℓ representation $\overline{\rho}$ of G, then $\mathrm{Hom}_k(m^\square/((m^\square)^2,\ell),k)$ naturally bijected with the 1-cocycles $G \to \mathrm{ad}(\overline{\rho})$.

And if $\overline{\rho}$ was absolutely irreducible with universal deformation ring R^{univ} then $\mathrm{Hom}_k(m^{\mathrm{univ}}/((m^{\mathrm{univ}})^2,\ell),k)$ naturally bijected with $H^1(G,\mathrm{ad}(\overline{\rho}))$.

Although I didn't mention it, there is a variant where you deform to $\mathcal{O}$-algebras ($\mathcal{O}$ the integers in a finite extension of $\mathbb{Q}_\ell$ with residue field k).

You just replace $(m/(m^2,\ell))$ with $(m/(m^2,m_{\mathcal{O}}))$ where $m_{\mathcal{O}}$ is the maximal ideal of $\mathcal{O}$.

All the proofs are the same.

Deformation rings are Noetherian

It is worth pointing out that an elementary (but not trivial) calculation in Lenstra–de Smit (Boston FLT volume) shows that a universal deformation ring R is Noetherian iff m_R/m_R^2 is finite-dimensional.

So all of our deformation and lifting rings are Noetherian, because the cohomology group $H^1(G_S, M)$ is finite for all finite M, although finiteness is a deep result.

(For example, it implies that there are only finitely many extensions of a number field of fixed degree and unramified outside a fixed set of primes).

In particular, all our deformation and lifting rings have finite Krull dimension.

Computing bounds for dimensions

Let's fix $\mathcal{O}$ the integers in a finite extension of $\mathbb{Q}_\ell$ with maximal ideal $m_\mathcal{O}$, and consider deformations/liftings to $\mathcal{O}$-algebras.

We can then go through the previous argument replacing $m^{\mathrm{univ}}/((m^{\mathrm{univ}})^2, \ell)$ with $m^{\mathrm{univ}}/((m^{\mathrm{univ}})^2, m_\mathcal{O})$.

The argument shows that if $\overline{\rho}$ is absolutely irreducible then the dual of the k-vector space $m^{\mathrm{univ}}/((m^{\mathrm{univ}})^2, m_\mathcal{O})$ naturally bijects with $H^1(G_S, \mathrm{ad}(\overline{\rho}))$.

Set $d_1 := \dim_k(H^1(G_S, \mathrm{ad}(\overline{\rho}))) < \infty$.

Choose a k-basis $\overline{x}_i$ of $m^{\mathrm{univ}}/((m^{\mathrm{univ}})^2, m_\mathcal{O})$, of size d_1.

One now checks that we have a surjection $\mathcal{O}[[x_1, x_2, \ldots, x_{d_1}]] \to R^{\mathrm{univ}}$.

In particular, the Krull dimension of R^{univ} is at most $d_1 + 1$.

So that's an upper bound – and here's how to get a lower bound.

Computing bounds for dimensions

Say R^{univ} is the universal deformation ring for $\overline{\rho}$.

Choose a surjection $\phi : \mathcal{O}[[x_1, x_2, \ldots, x_{d_1}]] \to R^{\text{univ}}$ and let J be its kernel.

I claim that $\mathrm{Hom}_k(J/(J^2, \mathfrak{m}_{\mathcal{O}}), k)$ naturally injects into $H^2(G_S, \mathrm{ad}(\overline{\rho}))$.

This is in Mazur's 1989 paper about deformation rings (Proposition 2), but the construction of the map is not hard.

We choose a basis for the universal deformation, giving $G \to \mathrm{GL}_n(R^{\text{univ}})$.

We then choose a set-theoretic inverse of ϕ giving a set-theoretic map $\sigma : G \to \mathrm{GL}_n(\mathcal{O}[[x_1, x_2, \ldots, x_{d_1}]])$.

Now we simply write down the function $G^2 \to \mathrm{GL}_n(\mathcal{O}[[x_1, x_2, \ldots, x_{d_1}]])$ sending (g, h) to $\sigma(gh)\sigma(g)^{-1}\sigma(h)^{-1}$ measuring its failure to be a group hom.

This reduces mod J to zero, so you get an induced 2-cocycle taking values in $(J/J^2) \otimes_k \mathrm{ad}(\overline{\rho})$.

You now push this forward along the map $J/J^2 \to k$.

Computing bounds for dimensions

We have $\phi : \mathcal{O}[[x_1, x_2, \ldots, x_{d_1}]] \to R^{\text{univ}}$ surjective with kernel J.

We just embedded the dual of the k-vector space $J/(J^2, \mathfrak{m}_{\mathcal{O}})$ into $H^2(G_S, \mathrm{ad}(\overline{\rho}))$.

The upshot is that there are at most $d_2 := \dim(H^2(G_S, \mathrm{ad}(\overline{\rho})))$ relations in the presentation $\phi : \mathcal{O}[[x_1, x_2, \ldots, x_{d_1}]] \to R^{\text{univ}}$.

Two consequences:

(1) If $H^2(G_S, \mathrm{ad}(\overline{\rho})) = 0$ then actually $R^{\text{univ}} \cong \mathcal{O}[[x_1, x_2, \ldots, x_{d_1}]]$ (because $J = 0$); and

(2) in general the Krull dimension of R^{univ} is at least $1 + d_1 - d_2$.

However the global Euler characteristic formula tells us that $\dim_k H^0(G_S, \mathrm{ad}(\overline{\rho})) - d_1 + d_2 = -\sum_{v|\infty} \dim_k(\mathrm{ad}(\overline{\rho})/\mathrm{ad}(\overline{\rho})^{G_v})$.

So a lower bound for the Krull dimension of R^{univ} is the easy-to-compute $1 + \dim_k H^0(G_S, \mathrm{ad}(\overline{\rho})) + \sum_{v|\infty} \dim_k(\mathrm{ad}(\overline{\rho})/\mathrm{ad}(\overline{\rho})^{G_v})$.

Modifications

In practice we don't care about R^{univ}. We care about deformations of G_S (which are automatically unramified outside S) with extra hypotheses.

The extra hypotheses were

(1) Cyclotomic determinant;
(2) Local conditions at $P \in S$ ("minimally ramified").

Enforcing cyclotomic determinant causes the cohomology group whose H^1 controls m/m^2 to become $\mathrm{ad}^0(\overline{\rho})$, the trace zero endomorphisms of k^2.

The reason: the dictionary between deformations and cocycles sent $\tilde{\rho} : G \to \mathrm{GL}_n(k[\epsilon])$ to the 1-cocycle c with $\tilde{\rho}(g) = \overline{\rho}(g)(\mathbf{1}_n + \epsilon c(g))$.

Enforcing $\det(\tilde{\rho}) = \det(\overline{\rho})$ thus means enforcing $\det(\mathbf{1}_n + \epsilon c(g)) = 1$.

But $\det(\mathbf{1}_n + \epsilon c(g)) = 1 + \epsilon\,\mathrm{tr}(c(g))$ so it enforces $\mathrm{tr}(c(g)) = 0$.

Local conditions

Recall: our S-minimal local conditions were of two kinds: if $P \mid \ell$ it was "flat at P" and if $P \nmid \ell$ it was "trace of $\rho(i)$ is 2 for any i in I_P."

When applied to deformations to $k[\epsilon]$ they all say "this cocycle lies in this subset of $H^1(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}(\overline{\rho}))$."

Our local conditions always satisfied three axioms.

These axioms ensured that the deformations remained representable when you added the conditions.

Indeed the axioms implied that adding the condition was the same as quotienting out the deformation ring by an ideal.

The axioms similarly imply that this subset of $H^1(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}(\overline{\rho}))$ where the axioms hold is a k-subspace.

This is exactly what $H^1_{\mathcal{L}}$ is all about.

Modifications

We'd like to get an understanding of the Krull dimension of rings like $R^{S-\mathrm{minl}}$.

So for $P \in S$ dividing ℓ let's define $L_P \subseteq H^1(F_P, \mathrm{ad}^0(\overline{\rho}))$ to be the subspace corresonding to flat deformations $\rho : \mathrm{Gal}(\overline{F}_P/F_P) \to \mathrm{GL}_2(k[\epsilon])$.

And for $P \in S$ prime to ℓ let's define $L_P \subseteq H^1(F_P, \mathrm{ad}^0(\overline{\rho}))$ to be the subspace corresponding to deformations $\rho : \mathrm{Gal}(\overline{F}_P/F_P) \to \mathrm{GL}_2(k[\epsilon])$ with trace 2 on all elements of I_P.

We then get a modified global cohomology group $H^1_{\mathcal{L}}(G_S, \mathrm{ad}^0(\overline{\rho}))$ which naturally bijects with the k-dual of $m/(m^2, m_{\mathcal{O}})$ for m the maximal ideal of $R^{S-\mathrm{minl}}$.

So we get a surjection $\mathcal{O}[[X_1, X_2, \ldots, X_d]] \to R^{S-\mathrm{minl}}$, where d is the dimension of $H^1_{\mathcal{L}}(G_S, \mathrm{ad}^0(\overline{\rho}))$.

The key idea

I've been whingeing that we can't control this global H^1 because it's like the ℓ-torsion in a class group.

But class groups are Galois groups (via the theory of the Hilbert class field).

The Cebotarev density theorem says that if E/F is a finite Galois extension of number fields, then every conjugacy class in the Galois group contains Frob_P for infinitely many primes P.

So by Cebotarev density we can at least say "I don't know what the size of the ℓ-torsion in this class group is, but we can at least pick a basis consisting of Frobenius elements".

We'll do an analogous thing here.

Taylor–Wiles primes

We then have these uncontrolled spaces $H^1_{\mathcal{L}}(G_S, \mathrm{ad}^0(\overline{\rho}))$ and $H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$.

We don't know much about their dimensions, but we have an easy formula for the difference of their dimension.

So the trick: let's increase S and $\mathcal{L}$ and make $H^1_{\mathcal{L}^\perp}$ vanish.

Say $0 \neq \phi \in H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$. Let's kill it.

Say $N \geq 1$ is an arbitrary positive integer.

Let's find a "Taylor–Wiles" prime v of F such that $v \notin S$, $|v| \equiv 1 \bmod \ell^N$, $\overline{\rho}(\mathrm{Frob}_v)$ has distinct eigenvalues in $\overline{k}$, and, most importantly, that $res_v(\phi) \in H^1(\mathrm{Gal}(\overline{F}_v/F_v), \mathrm{ad}^0(\overline{\rho})^*) \neq 0$.

Then we can add v to S, let $L_v^\perp := 0$ and throw this into $\mathcal{L}^\perp$, making the dimension of $H^1_{\mathcal{L}^\perp}$ go down. How do we find such a v?

Cebotarev density

Saying $|v| \equiv 1 \bmod \ell^N$ is a statement about a Frobenius element in a Galois group – it's "$\mathrm{Frob}_v = 1$ in $\mathrm{Gal}(F(\zeta_{\ell^N})/F)$".

Saying "$\overline{\rho}(\mathrm{Frob}_v)$ has distinct eigenvalues" is saying that Frob_v is in a certain subset of $\mathrm{Gal}(F^{\ker(\overline{\rho})}/F)$.

Saying $res_v(\phi)$ doesn't vanish in $H^1(F_v, \mathrm{ad}^0(\overline{\rho})^*)$ is a statement about Frob_v being in a certain subset of $\mathrm{Gal}(F^{\ker(\mathrm{ad}(\overline{\rho}))}/F)$.

The idea: v exists because of the Cebotarev density theorem.

But we have to ensure that these statements don't contradict each other.

There is some content here.

Furthermore I don't know a reference from the 1980s (even though all the ideas were known in the 1980s) so I shouldn't really skip this, but for reasons of time I'll have to.

Here's what goes into the argument.

Cebotarev application

We have a condition on Frob_v on $F(\zeta_\ell)$, a condition on $\overline{F}^{\ker(\overline{\rho})}$ and a condition on $\overline{F}^{\ker(\mathrm{ad}(\overline{\rho}))}$.

The first two conditions are compatible because we assume that $\overline{\rho}$ on $F(\zeta_\ell)$ is absolutely irreducible.

(for those who remember, this was OK in our applications because flatness at ℓ and the fact that our Galois representation was valued in $\mathrm{GL}_2(\mathbb{Z}/\ell\mathbb{Z})$ was enough.)

To check compatibility with the third, we need to make sure that $F(\zeta_\ell)$ isn't in the field cut out by the kernel of $\mathrm{ad}(\overline{\rho})$.

This is one of the few places we need $\ell \geq 5$.

Reminder: we can think of $\mathrm{ad}(\overline{\rho})$ as a Galois representation on $M_2(k)$, with g sending M to $\rho(g)M\rho(g)^{-1}$.

So $\mathrm{ad}(\overline{\rho})$ actually factors through $PGL_2(\overline{\mathbb{F}}_\ell)$.

Cebotarev application

To ensure that we can make $\mathrm{ad}(\overline{\rho})(\mathrm{Frob}_v)$ complicated but Frob_v trivial on $F(\zeta_{\ell N})$ we need to ensure that $F(\zeta_\ell)$ is not in the field cut out by $\mathrm{ad}(\overline{\rho})$.

But this field has a Galois group over F which embeds into $PGL_2(\overline{\mathbb{F}}_\ell)$.

And $\overline{\rho}$ is absolutely irreducible, so the image in PGL_2 is not cyclic.

So by the classification of finite subgroups of $PGL_2(\overline{\mathbb{F}}_\ell)$ it's dihedral, A_4, S_4, A_5, or $PSL_2(\mathbb{F})$ or $PGL_2(\mathbb{F})$ for some finite field $\mathbb{F}$.

And none of these groups have got a cyclic quotient of order $\ell - 1$ if $\ell \geq 5$.

But $F(\zeta_\ell)/F$ is cyclic of order $\ell - 1$ because ℓ is unramified in F.

Application

The upshot is that given any $N \geq 1$ and any $0 \neq \phi \in H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$ we can find a "Taylor–Wiles prime" of F, not in S, whose norm is 1 mod ℓ^N, with $\overline{\rho}(\mathrm{Frob}_v)$ having distinct eigenvalues, and such that $res_v(\phi) \neq 0$.

Once and for all let's let $s = s(\overline{\rho}, S)$ be the dimension of $H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$, with $\mathcal{L}$ the subspaces corresponding to deformations which are minimal at $P \in S$.

Applying the Taylor–Wiles construction inductively, we get a collection Q_N of s primes of F such that $H^1(G_S, \mathrm{ad}^0(\overline{\rho})^*) \to \oplus_{v \in Q_N} H^1(F_v, \mathrm{ad}^0(\overline{\rho})^*)$ is injective.

So if we define L_v for $v \in Q_N$ to be all of $H^1(F_v, \mathrm{ad}^0(\overline{\rho}))$ and throw these into $\mathcal{L}$, then $L_v^\perp = 0$ which will mean that $H^1_{\mathcal{L}^\perp}(G_{S \cup Q_N}, \mathrm{ad}^0(\overline{\rho})^*) = 0$.

So we now have a simple formula for the dimension of $H^1_{\mathcal{L}}(G_{S \cup Q_N}, \mathrm{ad}^0(\overline{\rho}))$ (because we have a simple formula for the difference $\dim_k(H^1_{\mathcal{L}}) - \dim_k(H^1_{\mathcal{L}^\perp})$).

Let's work it out.

Dimension of the cohomology group

We have s, the dimension of $H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$.

Here this $\mathcal{L}$ is the minimality conditions for primes in S.

Given a positive integer N we can find a set Q_N of primes with size s, such that if we extend $\mathcal{L}$ by $L_v = H^1(F_v, \mathrm{ad}^0(\overline{\rho}))$ and $L_v^\perp = 0$, we can make $H^1_{\mathcal{L}^\perp}(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho})^*) = 0$.

So the dimension of $H^1_{\mathcal{L}}(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho}))$ can be computed, and it's $\dim_k H^0(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho})) - \dim_k H^0(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho})^*) - \sum_{v|\infty} \dim_k \mathrm{ad}^0(\overline{\rho})^{G_v} + \sum_{P\in S\cup Q_N}(\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})))$.

Because $\overline{\rho}$ is irreducible, this is $0 - 0 - [F : \mathbb{Q}] + \sum_{P\in S\cup Q_N}(\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})))$.

But now we have to figure out what the dimension of these L_P spaces is.

Dimension of L_P

So far: $H^1_{\mathcal{L}}(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho}))$ has dimension
$-[F:\mathbb{Q}] + \sum_{P\in S\cup Q_N}(\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})))$.

The computation of $\dim_k L_P$ for $P \mid \ell$ was done in Ravi Ramakrishna's thesis and it's $[F_P:\mathbb{Q}_\ell] + \dim H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho}))$.

Because $\sum_{P\mid\ell}[F_P:\mathbb{Q}_\ell] = [F:\mathbb{Q}]$ these cancel and we're left with
$\sum_{P\in S\cup Q_N-\{\ell\}}(\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})))$.

For $P\in Q_N$, L_P is all of the H^1, so by local Euler charateristic the summand is $\dim_k H^2(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho}))$ which by local Tate duality is $\dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho}))$ (note that $|P| = 1 \bmod \ell$).

Now $\overline{\rho}$ sends Frob_P to a diagonal matrix with eigenvalues $\alpha\neq\beta$ (the Taylor–Wiles condition).

So $\mathrm{ad}^0(\overline{\rho})$ has eigenvalues $\alpha/\beta, 1, \beta/\alpha$ and only one of these is 1.

Conclusion: $\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})) = 1$ for $P\in Q_N$.

The story so far

$H^1_{\mathcal{L}}(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho}))$ has dimension
$\#Q_N + \sum_{P\in S-\{\ell\}}(\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})))$.

But $\#Q_N = s$, the dimension of $H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$.

So $\dim_k H^1_{\mathcal{L}}(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho})) =$
$s + \sum_{P\in S-\{\ell\}}(\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})))$.

Let's leave this for now and ask another question:

What deformation problem is $H^1_{\mathcal{L}}(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho}))$ related to?

It's deformations of $\overline{\rho}$ which are now allowed to ramify at the Taylor–Wiles primes in Q_N, and the local condition at these primes is nothing at all, because L_P was all of the local H^1 in those cases.

The story so far

We have been forced into thinking about a new deformation problem.

We start with F and S and $\overline{\rho}$ as usual.

We set $s := \dim_k H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$.

We have some random positive integer N whose role I have not made clear.

We have a set Q_N of s primes and are now considering deformations of $\overline{\rho}$ which are S-minimal (cyclo det, flat at ℓ, trace is 2 on inertia at the other $P \in S$) except that they are also allowed to ramify arbitrarily at $P \in Q_N$.

Let's call these Q_N-deformations, and say they're represented by R_{Q_N}.

Of course every S-minimal deformation is a Q_N-deformation, so we get maps $R_{Q_N} \to R^{S-\mathrm{minl}}$ in our category of compact Hausdorff local $\mathcal{O}$-algebras with residue field k.

General nonsense tells us that R_{Q_N} is topologically generated as an $\mathcal{O}$-algebra by $\dim_k H^1_{\mathcal{L}}(G_{S\cup Q_N}, \mathrm{ad}^0(\overline{\rho}))$ elements.

We just computed this dimension.

A numerical coincidence

We're allowing freedom of our deformations at a set Q_N of primes of size s.

We computed that the deformation ring was topologically generated by $s + \sum_{P \in S - \{\ell\}} (\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{F}_P/F_P), \mathrm{ad}^0(\overline{\rho})))$ elements.

If that messy extra sum were 0 then there would be a "numerical coincidence" that the set Q_N of "extra degrees of freedom" in our deformation problem had the same size as the number of generators of R_{Q_N}.

Sadly, this is typically not the case.

So let's temporarily *make* it the case, by *assuming* that S *only* contains primes above ℓ.

Then the sum is a sum over the empty set and the numerical coincidence occurs.

The case when S only contains primes above ℓ is called the "minimal case".

Let's finish the proof in this case, and then talk about the general case.

Let's prove this theorem

Let's prove the automorphy lifting theorem in the minimal case.

We have F totally real of even degree, $\ell \geq 5$ prime, k a finite field of characteristic ℓ, S the places of F above ℓ.

We have $\overline{\rho} : G_S \to \mathrm{GL}_2(k)$ flat at ℓ, cyclotomic character.

We assume $\overline{\rho}$ comes from an automorphic form of level $U_0(1)$.

If $R = R^{S-\mathrm{minl}}$ is the ring representing S-minimal deformations of $\overline{\rho}$ (unramified outside S, flat at S, cyclotomic det), then our assumption gives us a nontrivial map $R \to \overline{\mathbb{Q}}_\ell$.

What we want is that R is $\mathbb{Z}_\ell$-finite, and that *every* map $R \to \overline{\mathbb{Q}}_\ell$ comes from an automorphic form.

Modular deformations

We have this finite-dimensional $\mathbb{Q}_\ell$-vector space $\mathcal{A}_D(U_0(1);\mathbb{Q}_\ell)$ equipped with the actions of the "good" Hecke operators T_P for $P \notin S$.

Standard facts from the theory of automorphic forms tell us that the T_P are simultaneously diagonalisable over $\overline{\mathbb{Q}}_\ell$.

Let $\mathcal{A}_D(U_0(1);\overline{\mathbb{Q}}_\ell)_{\overline{\rho}}$ denote the subspace of $\mathcal{A}_D(U_0(1);\overline{\mathbb{Q}}_\ell)$ spanned by the eigenforms ϕ such that the associated Galois representations ρ_ϕ have the following two properties:

First, they have cyclotomic determinant (equivalently: are constant on cosets of $(\mathbb{A}_F^\infty)^\times$).

Secondly, their mod ℓ reduction is isomorphic to $\overline{\rho}$.

These are the only automorphic forms we care about.

Deep local-global theorems (which I stated previously) say that ρ_ϕ is S-minimal.

Modular deformations

Choose a sufficiently large finite extension $K/\mathbb{Q}_\ell$ so that all of the eigenvalues of all of the T_P on $\mathcal{A}_D(U_0(1);\overline{\mathbb{Q}}_\ell)$ are in K.

Because the T_P are also endomorphisms of the lattice $\mathcal{A}_D(U_0(1);\mathcal{O}) \subseteq \mathcal{A}_D(U_0(1);K)$, the eigenvalues are integral and hence in $\mathcal{O}$.

We have $\mathcal{A}_D(U_0(1);\overline{\mathbb{Q}}_\ell)_{\overline{\rho}}$, the finite-dimensional $\overline{\mathbb{Q}}_\ell$-vector space of automorphic forms we care about.

Let's define $\mathcal{A}_D(U_0(1);K)_{\overline{\rho}}$ to be $\mathcal{A}_D(U_0(1);\overline{\mathbb{Q}}_\ell)_{\overline{\rho}} \cap \mathcal{A}_D(U_0(1);K)$.

Let $\mathbb{T}$ denote the (commutative) $\mathcal{O}$-algebra of endomorphisms of $\mathcal{A}_D(U_0(1);K)_{\overline{\rho}}$ generated by the T_P.

If we choose a basis of eigenforms then $\mathbb{T}$ is just a subalgebra of the diagonal matrices in a matrix ring.

Modular deformations

We have the K-space $\mathcal{A}_D(U_0(1);K)_{\overline{\rho}}$ containing all systems of eigenvalues we care about.

Write this as a sum $\oplus_{i=1}^n E_i$ of $\mathbb{T}$-eigenspaces.

Each eigenspace gives us $\rho_i : G_S \to \mathrm{GL}_2(K)$ with $\mathtt{tr}(\rho_i(\mathrm{Frob}_P))$ equal to the eigenvalue of T_P for $P \notin S$.

Compactness of G_S means that the image of ρ_i can be conjugated into $\mathrm{GL}_2(\mathcal{O})$, so let's assume this.

Each eigenspace E_i gives a ring homomorphism $\mathbb{T} \to \mathcal{O}$, sending a Hecke operator to its E_i-eigenvalue.

So we get a ring homomorphism $\mathbb{T} \to \oplus_{i=1}^n \mathcal{O}$ which is easily checked to be injective.

Identify $\mathbb{T}$ with its image in $\oplus_{i=1}^n \mathcal{O}$.

Modular deformations

Each eigenspace gives us $\rho_i : G_S \to \mathrm{GL}_2(\mathcal{O})$ with $\mathrm{tr}(\rho_i(\mathrm{Frob}_P))$ equal to the image of T_P in the corresponding $\mathcal{O}$.

So their sum is $\rho : G_S \to \mathrm{GL}_2(\oplus_{i=1}^n \mathcal{O})$ with $\mathrm{tr}(\rho(\mathrm{Frob}_P)) = T_P \in \mathbb{T} \subseteq \oplus_{i=1}^n \mathcal{O}$.

General facts about representations into local rings whose reductions are absolutely irreducible show that ρ furthermore descends to a map $G_S \to \mathrm{GL}_2(\mathbb{T})$.

(the point is that the trace of $\rho(\mathrm{Frob}_P)$ is in $\mathbb{T}$ for all good P, and these are dense in the Galois group, so all traces land in $\mathbb{T}$ and now it's algebra.)

So we get a map $\rho : G_S \to \mathrm{GL}_2(\mathbb{T})$ and it's easily checked to be S-minimal.

This gives us a map $R \to \mathbb{T}$.

Note that $\mathbb{T} \subseteq \mathcal{O}^n$ so has no nilpotents.

If we can prove that this induces an isomorphism from R modulo its nilpotent ideal to $\mathbb{T}$ then our main theorem is an easy corollary. Let's just check this.

What we want

The theorem we're trying to prove is that R is finitely-generated as a $\mathbb{Z}_\ell$-module, and that all maps $R \to \overline{\mathbb{Q}}_\ell$ induce representations $G_S \to \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$ which come from automorphic forms.

If we can prove that $R \to \mathbb{T}$ induces an isomorphism $R^{red} \to \mathbb{T}$ then because $\mathbb{T}$ is a finitely-generated $\mathbb{Z}_\ell$-module and R is Noetherian, R is also a finitely-generated $\mathbb{Z}_\ell$-module.

And because any map $R \to \overline{\mathbb{Q}}_\ell$ must be zero on nilpotents, it must factor through $\mathbb{T}$ and thus through one of the $\mathcal{O}$, meaning that it's associated to the corresponding eigenspace E_i.

Upshot: we have a map $R \to \mathbb{T}$ and we want to show that it induces an isomorphism $R^{red} \to \mathbb{T}$.

We'll now introduce more R's and $\mathbb{T}$'s, so let me rename these rings $R_\emptyset$ and $\mathbb{T}_\emptyset$.

Enter the Taylor–Wiles primes

Recall $s = \dim_k H^1_{\mathcal{L}^\perp}(G_S, \mathrm{ad}^0(\overline{\rho})^*)$ is some natural number depending only on $\overline{\rho}$ and S.

Now let N be a positive integer.

Under our simplified assumption that $S = \{P \mid \ell\}$ we showed that there was a set of Taylor–Wiles primes Q_N of size s such that if R_{Q_N} was deformations unramified outside ℓQ_N and flat at ℓ, then this deformation ring is topologically generated by s elements.

It would be nice to have a $\mathbb{T}_{Q_N}$ and we make this by acting on automorphic forms of level $U_1(Q_N)$, where the local component at $P \in Q_N$ is that your matrices are of the form $\begin{pmatrix} a & b \\ \varpi_P c & d \end{pmatrix}$ with a/d mod ϖ_P in $(\mathcal{O}_F/P)^\times$ having order prime to ℓ.

Recall that $\#(\mathcal{O}_F/P)$ is 1 mod ℓ^N (this is where N comes in) so this is quite a bit smaller than $U_0(Q_N)$.

Higher level R and T

We can consider the space $S_{Q_N} := \mathcal{A}_D(U_1(Q_N); K)_{\overline{\rho}}$ with its action of the Hecke algebra $\mathbb{T}_{Q_N}$ generated over $\mathcal{O}$ by T_P for $P \notin S \cup Q_N$ and a certain operator U_P for each $P \in Q_N$.

This is a bigger space of forms, corresponding to the fact that we have more freedom in our deformations.

As before, we get a deformation $\rho_{Q_N} : G_{S \cup Q_N} \to \mathrm{GL}_2(\mathbb{T}_{Q_N})$ and hence a map $R_{Q_N} \to \mathbb{T}_{Q_N}$.

So we get a picture

$$\begin{tikzcd}
R_{Q_N} \arrow[r, two heads] \arrow[d, two heads] & \mathbb{T}_{Q_N} \arrow[d, two heads] & \curvearrowright & S_{Q_N} \\
R_{\emptyset} \arrow[r, two heads] & \mathbb{T}_{\emptyset} & \curvearrowright & S_{\emptyset} \arrow[u, hook]
\end{tikzcd}$$

(the maps between the rings are surjective because deformation rings are generated by traces, and traces of Frobenius elements are Hecke operators).

The algebra we need

$$\begin{tikzcd} R_{Q_N} \arrow[r, two heads] \arrow[d, two heads] & \mathbb{T}_{Q_N} \arrow[d, two heads] & \curvearrowright & S_{Q_N} \\ R_\emptyset \arrow[r, two heads] & \mathbb{T}_\emptyset & \curvearrowright & S_\emptyset \arrow[u, hook] \end{tikzcd}$$

We need to prove that the kernel of this bottom map is the nilpotent elements of $R_\emptyset$.

The other things we know: R_{Q_N} is topologically generated by s elements, so there's a surjection $R_\infty := \mathcal{O}[[X_1, X_2, \ldots, X_s]] \to R_{Q_N}$.

The last piece of the puzzle is understanding this group $U_1(Q_N)$, which was a normal subgroup of $U_0(Q_N)$ consisting of elements which locally at $P \in Q_N$ are of the form $\begin{pmatrix} a & b \\ \varpi_P c & d \end{pmatrix}$ with a/d mod P having order prime to ℓ.

Δ_{Q_N}

We have $U_0(Q_N)$ which locally at $P \in Q_N$ is matrices which mod P are upper triangular.

And then $U_1(Q_N)$ is matrices which locally mod P are upper triangular with a/d of order prime to ℓ.

So $\Delta_{Q_N} := U_0(Q_N)/U_1(Q_N)$ is a product of the Sylow ℓ-subgroups of $(\mathcal{O}_F/P)^\times$ for P running through Q_N.

Note that $(\mathcal{O}_F/P)^\times$ has size a multiple of ℓ^N; this is where N comes in.

This group Δ_{Q_N} acts on *everything* ("by functoriality").

Example: it acts on S_{Q_N} (right translating $f : D^\times \backslash D_f^\times / U_1(Q_N) \to X$ by an element of $U_0(Q_N)$ is still $U_1(Q_N)$-invariant.)

In fact, one checks that if $S_\emptyset$ is $\mathcal{O}$-free of rank r then S_{Q_N} is $\mathcal{O}[\Delta_{Q_N}]$-free of the same rank.

In fact Δ_{Q_N} acts on S_{Q_N} and $S_\emptyset$ is the coinvariants.

More on Δ_{Q_N}.

In fact R_{Q_N} is naturally a $\mathcal{O}[\Delta_{Q_N}]$-module.

A deformation of $\overline{\rho}$ locally at $P \in Q_N$ must be the sum of two characters (because $|P| = 1 \bmod \ell$ and $\overline{\rho}(\mathrm{Frob}_P)$ has two distinct eigenvalues).

The characters must be tame (as $P \nmid \ell$) so give (via local class field theory) characters of $(\mathcal{O}_F/P)^\times$.

For each $P \in Q_N$ we choose a "favourite" eigenvalue and thus a favourite character.

This makes R_{Q_N} into an $\mathcal{O}[\Delta_{Q_N}]$-algebra.

Of course the Sylow ℓ-subgroup of $(\mathcal{O}_F/P)^\times$ is cyclic; choose a generator for each $P \in Q_N$.

Then we get the following beefed-up diagram.

Our diagram

$$\begin{tikzcd}
\Lambda := \mathcal{O}[[\mathbb{Z}_\ell^s]] \cong \mathcal{O}[[Y_1, Y_2, \ldots, Y_s]] \arrow[r, two heads, "\alpha"] \arrow[d, "\beta"'] & \mathcal{O}[\Delta_{Q_N}] \arrow[d] & & \\
R_\infty := \mathcal{O}[[X_1, X_2, \ldots, X_s]] \arrow[r, two heads, "\gamma"] & R_{Q_N} \arrow[r, two heads] \arrow[d, two heads] & \mathbb{T}_{Q_N} \arrow[d, two heads] & \curvearrowright S_{Q_N} \\
 & R_\emptyset \arrow[r, two heads] & \mathbb{T}_\emptyset & \curvearrowright S_\emptyset
\end{tikzcd}$$

Here α is a map sending each $1 \in \mathbb{Z}_\ell$ to a generator of the cyclic Sylow ℓ-subgroup of $(\mathcal{O}_F/P)^\times$ for $P \in Q_N$.

γ is a random map sending the X_i to generators of $m/(m^2, m_\mathcal{O})$.

The numerical coincidence is that Λ and R_∞ both have s generators.

β is a random map just made to make the diagram commute.

Now I claim it's just commutative algebra.

Koenig's Lemma

One weird thing is that these diagrams (one for each N) are completely unrelated.

But we are going to "take their limit" anyway.

The ring $\Lambda \cong \mathcal{O}[[Y_1, Y_2, \ldots, Y_s]]$ is local with maximal ideal m_Λ, and mapped to every other ring in the diagram.

We had these maps $\Lambda \to \mathcal{O}[\Delta_{Q_N}]$ and as N gets bigger the Δ_{Q_N} get bigger so the kernels of these maps get smaller.

So if J is a fixed open ideal of Λ, eventually the kernels of all these maps will be subsets of J.

Because S_{Q_N} is $\mathcal{O}[\Delta_{Q_N}]$-free of fixed rank (the $\mathcal{O}$-rank r of $S_\emptyset$) we see that the S_{Q_N}/J will have constant size for N sufficiently large (they will ultimately be isomorphic to $(\Lambda/J)^r$).

So for a fixed open ideal $J \subseteq \Lambda$ we have infinitely many $R_\infty := \mathcal{O}[[X_1, \ldots, X_s]]$-modules S_{Q_N}/J with a global bound $\#(\Lambda/J)^r$ on their finite sizes.

Koenig's Lemma

If $J' \subseteq J$ then we get R_∞-module maps $S_{Q_N}/J' \to S_{Q_N}/J$.

So choose a decreasing sequence of open ideals $J_1 \supset J_2 \supset J_3 \supset \cdots$ of Λ with zero intersection.

For J_1 choose an infinite subset $N_{1,1} < N_{1,2} < N_{1,3} < \cdots$ of naturals such that $S_{Q_{N_{1,i}}}/J_1$ are all isomorphic R_∞-modules.

Now bank $N_{1,1}$ and choose an infinite subset $N_{2,2} < N_{2,3} < N_{2,4} < \cdots$ of $\{N_{1,2}, N_{1,3}, N_{1,4}, \ldots\}$ such that the $S_{Q_{N_{2,i}}}/J_2$ are all isomorphic R_∞-modules.

Now bank $N_{2,2}$ and continue.

We get a compatible sequence $S_{Q_{N_{k,k}}}/J_k$ of R_∞-modules, so we can take the projective limit to get R_∞ acting on some random space S_∞ which is now not really a space of modular forms at all.

It has been glued together from spaces of modular forms of unrelated levels, and is an R_∞-module via some unnatural action which involved choosing a random presentation of R_{Q_N} because its tangent space was s-dimensional.

New picture

$$
\begin{tikzcd}
\Lambda \cong \mathcal{O}[[Y_1, Y_2, \ldots, Y_s]] \arrow[r, two heads] \arrow[d] & R_\infty \cong \mathcal{O}[[X_1, X_2, \ldots, X_s]] \arrow[d, two heads] & \curvearrowright S_\infty \\
\Lambda/(Y_1, \ldots, Y_s) \arrow[r, two heads] & R_\emptyset \arrow[r, two heads] & \mathbb{T}_\emptyset \curvearrowright S_\emptyset
\end{tikzcd}
$$

Each S_{Q_N} was free over $\mathcal{O}[\Delta_{Q_N}]$ of some fixed rank r. So $S_{Q_{N_{k,k}}}/J_k$ is free rank r over Λ/J_k.

So S_∞ is free over Λ of rank r.

Recall that the *depth* of an R_∞-module M is the maximum length of an M-regular sequence in the maximal ideal of R_∞.

Here a sequence $x_1, x_2, \ldots, x_n$ is an M-regular sequence if x_i is not a zero divisor in $M/(x_1, x_2, \ldots, x_{i-1})$ for all i and $M/(x_1, x_2, \ldots, x_n) \neq 0$.

So $\mathrm{depth}_\Lambda(S_\infty) = \mathrm{depth}_\Lambda(\Lambda)$.

And $\mathrm{depth}_\Lambda(\Lambda) = \dim(\Lambda)$ as Λ is regular.

The calculation

We have $\Lambda \cong \mathcal{O}[[Y_1, \ldots, Y_s]] \to R_\infty$ acting on S_∞.

We have $s + 1 = \mathrm{depth}_\Lambda(S_\infty)$.

This is $\leq \mathrm{depth}_{R_\infty}(S_\infty)$ (obvious).

This is $\leq \dim(R_\infty / \mathrm{Ann}_{R_\infty}(S_\infty))$ (standard fact: depth is at most dimension).

And this is of course $\leq \dim(R_\infty)$.

Which is $s + 1$.

So (by the numerical coincidence) all these $\leq$s are equalities.

So $\mathrm{Ann}_{R_\infty}(S_\infty) = 0$ (because R_∞ is an integral domain.)

But $S_\emptyset$ is just $S_\infty/(Y_1, \ldots, Y_s)$, so $\mathrm{Supp}_{R_\infty}(S_\emptyset) = \mathrm{Spec}(R_\infty/(Y_1, \ldots, Y_s))$, which contains all of $\mathrm{Spec}(R_\emptyset)$ as the surjection $R_\infty \to R_\emptyset$ factors through $R_\infty/(Y_1, \ldots, Y_s)$.

So $\mathrm{Supp}_{R_\emptyset}(S_\emptyset) = \mathrm{Spec}(R_\emptyset)$, and hence $\mathrm{Ann}_{R_\emptyset}(S_\emptyset)$ is nilpotent.

Thus the kernel of $R_\emptyset \to \mathbb{T}_\emptyset$ is nilpotent and this is what we wanted.

The general case

I have cut a corner here.

I assumed that S contained *only* the primes above ℓ.

In the general case $\overline{\rho}$ is automorphic of some level bigger than 1.

So we have an extra set S_r of primes coprime to ℓ where ρ, the automorphic lift of $\overline{\rho}$, may ramify.

When we allow these primes to ramify, we get a contribution of 3 for each of them, and the numerical coincidence doesn't work.

We still have $\Lambda = \mathcal{O}[[Y_1, \ldots, Y_s]]$ but the R_{Q_N} are generated over $\mathcal{O}$ by $s + 3\#S_r$ elements.

Originally (in the Wiles paper) this non-minimal case was dealt with by carefully measuring how $R_\emptyset$ and $T_\emptyset$ changed as you allowed more primes to ramify.

The trick (discovered by Kisin) that we'll formalize in the FLT project is to simply modify the argument in the minimal case.

But how do we account for this extra factor of $3\#S_r$?

Kisin's trick

The trick is that each of the local lifting rings R_P^{loc} (parametrising local lifts of $\overline{\rho}|\operatorname{Gal}(\overline{F}_P/F_P)$ for P a prime where our modular lift ρ of $\overline{\rho}$ ramifies, subject to the usual condition $\mathtt{tr}(\rho(i)) = 2$) can be explicitly computed.

These local rings have generic fibres which are 3-dimensional.

So one repeats the argument but with $\mathcal{O}$ replaced by the completed tensor product $R^{\mathrm{loc}} := \hat{\bigotimes}_{P \in S_r} R_P^{\mathrm{loc}}$.

The numerical coincidence is recovered, and the commutative algebra I presented above *nearly* goes through exactly the same.

There is just one glitch.

Fixing the glitch

I have skipped so many details in this lecture that we may as well skip one more.

But this is a rather delicate point, and was certainly not known in the 1980s.

The now-fixed numerical coincidence implies that $\mathrm{Supp}_{R_\infty}(S_\infty)$ is a union of irreducible components of $\mathrm{Spec}(R_\infty)$ but R_∞ is no longer an integral domain.

Instead of being $\mathcal{O}[[X_1, X_2, \ldots, X_s]]$ it is $R^{\mathrm{loc}}[[X_1, X_2, \ldots, X_s]]$.

We need that $\mathrm{Supp}_{R_\infty}(S_\infty)$ is all of these components in order to prove that the kernel of $R_\emptyset \to \mathbb{T}_\emptyset$ is nilpotent.

The irreducible components of R_∞ biject with the irreducible components of R^{loc}, which is the product of the irreducible components of the R_P^{loc}.

Unfortunately the explicit computation of R_P^{loc} shows that it has two components, not one :-(

)

Fixing the glitch

The issue is that we base changed $\overline{\rho}$ so that it was trivial at $P \in S_r$, so there are two kinds of deformations to characteristic zero:

Unramified deformations, and ramified but minimally ramified deformations.

These give us two irreducible components of $R_P^{\mathrm{loc}}[1/\ell]$ which reduce mod ℓ to two irreducible components of R_P^{loc}.

Could it be that entire components of $R_\emptyset$ are not modular?

Equivalently, could it be that entire components of $R_\emptyset/\ell$ are not in the image of the map from $\mathrm{Spec}(\mathbb{T}_\emptyset/\ell)$?

Kisin proves this can't happen, by looking at deformations of $\overline{\rho}$ such that a fixed generator of tame inertia at $P \in S_r$ has char poly $(X-\zeta_\ell)(X-\zeta_\ell^{-1})$ instead of $(X-1)^2$.

Fixing the glitch

The point is that in characteristic ℓ, the two deformation problems coincide, because $(X - \zeta_\ell)(X - \zeta_\ell^{-1})$ and $(X - 1)^2$ reduce to the same thing mod $m_{\mathcal{O}}$.

So the same numerical coincidences all work.

But on the generic fibre the story is different.

The corresponding local deformation rings $R_P^{\mathrm{loc}'}$ have one irreducible component generically, but two on the special fibre (think $\mathcal{O}[[X, Y]]/(XY - \ell)$).

The numerical coincidence shows that all these components of the generic fibre are modular, so all components of the special fibre are modular.

So all components of the generic fibre of our original deformation problem are modular, and this finishes the argument.

And with that, I will declare myself as having staggered over the finish line. Thanks for coming!